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NEET · PHYSICS · CHAPTER-WISE PYQS

Oscillations

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	57 questions
Answer key	at the end

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MCQ (Single Correct Answer) 57 questions

Q1. RE-NEET 2026

A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period $2T$. The value of ρ_2/ρ_1 is:

Open solution

(A) 1/4 (B) 4 (C) 2 (D) $1/2$

Q2. RE-NEET 2026

Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is $m\,\mathrm{kg}$ and the spring constant is $k\mathrm{Nm}^{-1}$. At a given instant, the extension of the spring is x-meter and the speed of the particle is $v\,\mathrm{ms}^{-1}$. On the $x-v$ plane, if the graph of v as a function of x is a circle, then the correct option is:

Open solution

(A) $k=\sqrt{m}$ (B) $k=\frac{1}{m}$ (C) $k=m$ (D) $k=m^2$

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Q3. NEET 2026

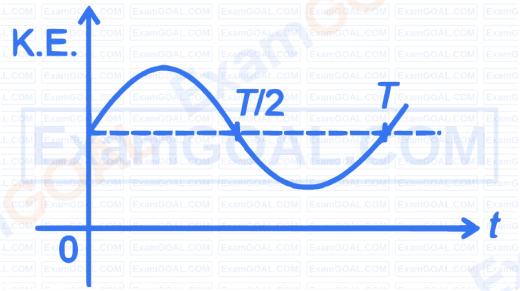
For a simple pendulum, having time period T, the variation of kinetic energy (K.E.) with time (t) is represented by:

Open solution

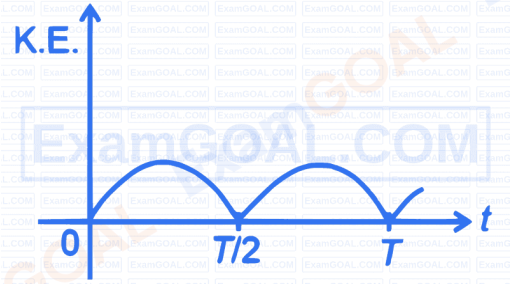
(A)



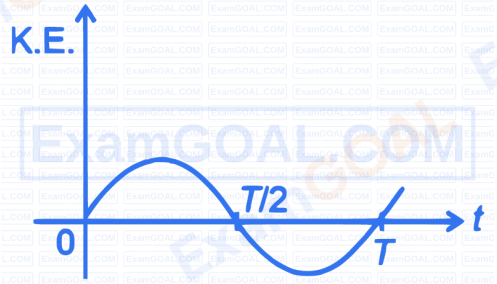
(B)



(C)



(D)



Q4. NEET 2026

The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately:

(Consider mass of the bob $= 20\,\text{g}$)

Open solution

(A) $0.2\,\text{m/s}$ (B) $1.41\,\text{m/s}$ (C) $14.1\,\text{m/s}$ (D) $2.0\,\text{m/s}$

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Q5. NEET 2026

Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as: (Take $\pi^2 = 9.8$, and $g = 9.8\ \mathrm{m/s^2}$)

Open solution

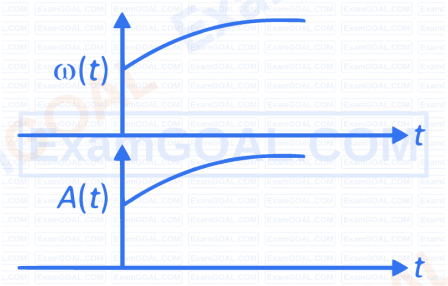
(A) 0.75 m (B) 1.5 m (C) 2 m (D) 1 m

Q6. NEET 2025

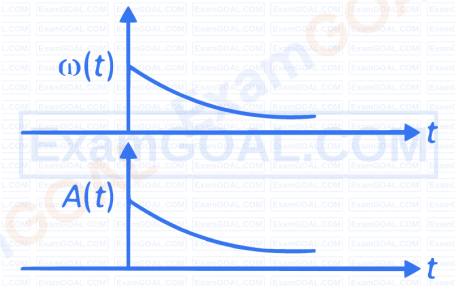
In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t. Which one of the following options schematically depicts these changes correctly?

Open solution

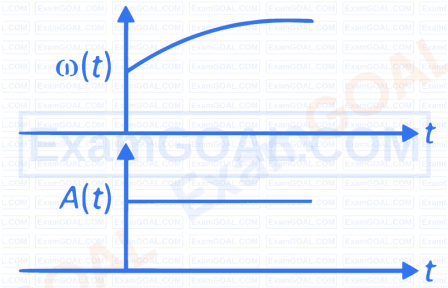
(A)



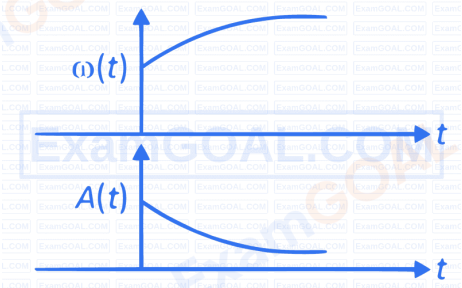
(B)



(C)



(D)



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Q7. NEET 2025

Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2, respectively, oscillate vertically. If their maximum speeds are the same, the ratio (AQ/A_ρ) of the amplitude AQ of mass Q to the amplitude A_ρ of mass P is

(A) $\sqrt{\frac{k_2}{k_1}}$ (B) $\sqrt{\frac{k_1}{k_2}}$ (C) $\frac{k_2}{k_1}$ (D) $\frac{k_1}{k_2}$

Open solution

Q8. NEET 2024 (Re-Examination)

A particle executing simple harmonic motion with amplitude A has the same potential and kinetic energies at the displacement

(A) $2\sqrt{A}$ (B) $\frac{A}{2}$ (C) $\frac{A}{\sqrt{2}}$ (D) $A\sqrt{2}$

Open solution

Q9. NEET 2024 (Re-Examination)

The two-dimensional motion of a particle, described by $\vec{r} = (\hat{i} + 2\hat{j})A\cos\omega t$ is a/an:

A. parabolic path

B. elliptical path

C. periodic motion

D. simple harmonic motion

Choose the correct answer from the options given below:

(A) B, C and D only (B) A, B and C only (C) A, C and D only (D) C and D only

Open solution

Q10. NEET 2024

If $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)\text{m}$ represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are

(A) 5 cm, 2 s (B) 5 m, 2 s (C) 5 cm, 1 s (D) 5 m, 1 s

Open solution

Q11. NEET 2024

If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

(A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) $2\sqrt{3}$ (D) 4

Open solution

Q12. NEET 2023 Manipur

A simple pendulum oscillating in air has a period of $\sqrt{3}\,\text{s}$. If it is completely immersed in non-viscous liquid, having density $\left(\frac{1}{4}\right)^{\text{th}}$ of the material of the bob, the new period will be :-

(A) $2\sqrt{3}\,\text{s}$ (B) $\frac{2}{\sqrt{3}}\,\text{s}$ (C) $2\,\text{s}$ (D) $\frac{\sqrt{3}}{2}\,\text{s}$

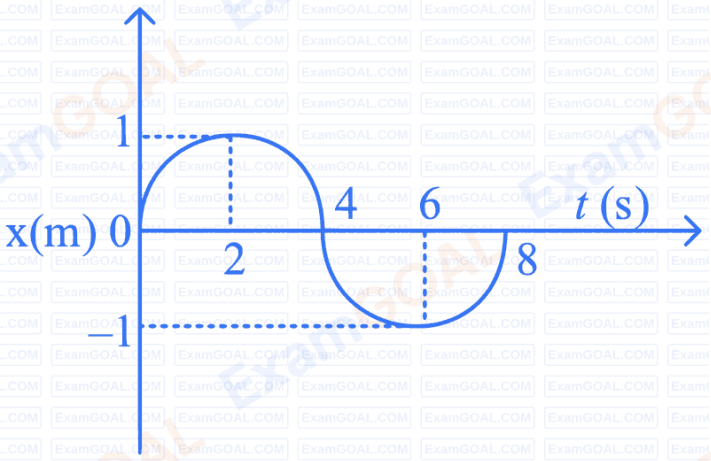
Open solution

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Q13. NEET 2023

Open solution

The x - t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2\,\text{s}$ is :



(A) $-\frac{\pi^2}{8}\,\text{ms}^{-2}$

(B) $\frac{\pi^2}{16}\,\text{ms}^{-2}$

(C) $-\frac{\pi^2}{16}\,\text{ms}^{-2}$

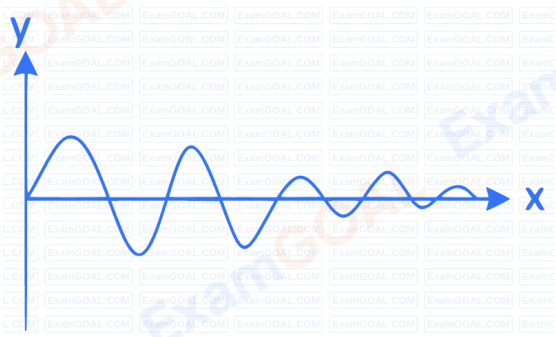
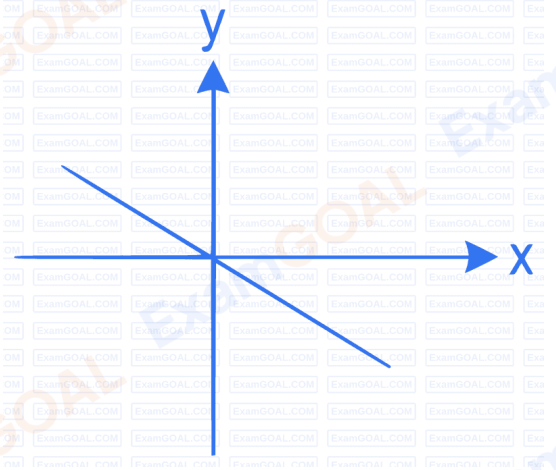
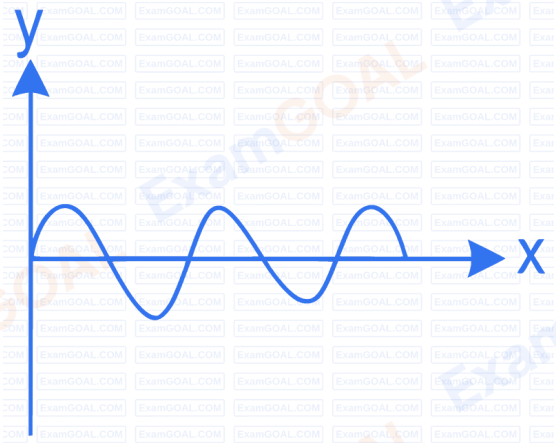
(D) $\frac{\pi^2}{8}\,\text{ms}^{-2}$

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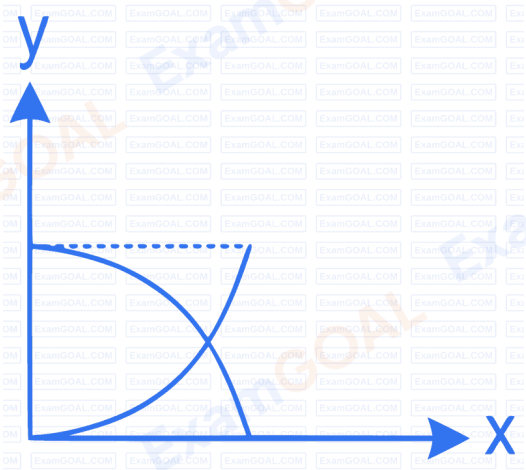
Q14. NEET 2022 Phase 2

Open solution

Match List-I with List-II

	List-I (x-y graphs)		List-II (Situations)
(a)		(i)	Total mechanical energy is conserved
(b)		(ii)	Bob of a pendulum is oscillating under negligible air friction
(c)		(iii)	Restoring force of a spring

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	List-I (x-y graphs)		List-II (Situations)
(d)		(iv)	Bob of a pendulum is oscillating along with air friction

Choose the correct answer from the options given below

(A) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(B) (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

(C) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(D) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

Q15. NEET 2022 Phase 2

Identify the function which represents a non-periodic motion.

(A) $\sin(\omega t + \pi/4)$ (B) $e^{-\omega t}$ (C) $\sin \omega t$ (D) $\sin \omega t + \cos \omega t$

Open solution

Q16. NEET 2022 Phase 1

Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is :

(A) 11 (B) 9 (C) 10 (D) 8

Open solution

Q17. NEET 2021

A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is :

(A) 4n (B) n (C) 2n (D) 3n

Open solution

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Q18. NEET 2021

A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is -

(A) 0.628 s (B) 0.0628 s (C) 6.28 s (D) 3.14 s

Open solution

Q19. NEET 2020 Phase 1

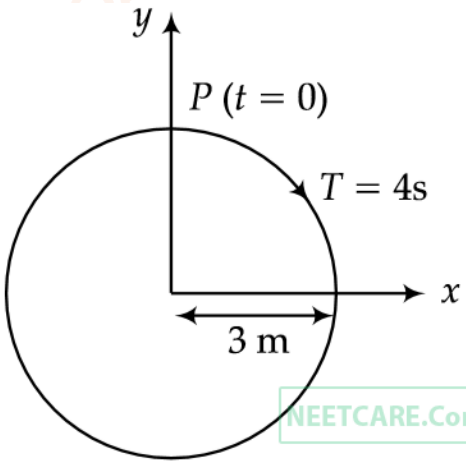
The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :

(A) $\frac{3\pi}{2} rad$ (B) $\frac{\pi}{2} rad$ (C) zero (D) π rad

Open solution

Q20. NEET 2019

The radius of circle, the period of revolution, initial position and sense of revolution are indicated is the figure.



NEETCARE.Com

Open solution

y- projection of the radius vector of rotating particle P is :

(A) y(t) = 4sin$\left(\frac{\pi t}{2}\right)$, where y in m

(B) y(t) = 3cos$\left(\frac{3\pi t}{2}\right)$, where y in m

(C) y(t) = 3cos$\left(\frac{\pi t}{2}\right)$, where y in m

(D) y(t) = -3cos2πt, where y in m

Q21. NEET 2019

Average velocity of a particle executing SHM in one complete vibration is :

(A) Aω (B) $\frac{A\omega^2}{2}$ (C) $\frac{A\omega}{2}$ (D) zero

Open solution

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Q22. NEET 2019

The displacement of a particle executing simple harmonic motion is given by

$y = A_0 + A \sin\omega t + B \cos\omega t$.

Then the amplitude of its oscillation is given by :

(A) $\sqrt{A^2 + B^2}$

(B) A + B

(C) A + $\sqrt{A^2 + B^2}$

(D) $\sqrt{A_0^2 + (A + B)^2}$

Open solution

Q23. NEET 2018

A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m s^{-2} at a distance of 5 m from the mean position. The time period of oscillation is

(A) 2π s

(B) π s

(C) 2 s

(D) 1 s

Open solution

Q24. NEET 2017

A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constant is K'. Then they are connected in parallel and force constant is k". Then k' : k" is

(A) 1 : 9

(B) 1 : 11

(C) 1 : 14

(D) 1 : 6

Open solution

Q25. NEET 2017

A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in second is

(A) $\frac{\sqrt{5}}{2\pi}$

(B) $\frac{4\pi}{\sqrt{5}}$

(C) $\frac{2\pi}{\sqrt{3}}$

(D) $\frac{\sqrt{5}}{\pi}$

Open solution

Q26. NEET 2016 Phase 2

A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is

(A) $\frac{3}{4}$

(B) $\frac{4}{3}$

(C) $\frac{16}{9}$

(D) $\frac{9}{16}$

Open solution

Q27. AIPMT 2015

A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be

(A) $\frac{\beta^2}{\alpha}$

(B) $\frac{2\pi\beta}{\alpha}$

(C) $\frac{\beta^2}{\alpha^2}$

(D) $\frac{\alpha}{\beta}$

Open solution

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Q28. AIPMT 2015 Cancelled Paper

A particle is executing SHM along a straight line. Its velocities at distances x_1 and x_2 from the mean position are V_1 and V_2 respectively. Its time period is

Open solution

(A) $2\pi\sqrt{\frac{V_1^2+V_2^2}{x_1^2+x_2^2}}$

(B) $2\pi\sqrt{\frac{V_1^2-V_2^2}{x_1^2-x_2^2}}$

(C) $2\pi\sqrt{\frac{x_1^2+x_2^2}{V_1^2+V_2^2}}$

(D) $2\pi\sqrt{\frac{x_2^2-x_1^2}{V_1^2-V_2^2}}$

Q29. AIPMT 2015 Cancelled Paper

When two displacements represented by y_1 = a $\sin(\omega t)$ and y_2 = b $\cos(\omega t)$ aresuperimposed the motion is

Open solution

(A) simple harmonic with amplitude $\sqrt{a^2+b^2}$

(B) simple harmonic with amplitude $\frac{(a+b)}{2}$

(C) not a simple harmonic

(D) simple harmonic with amplitude $\frac{a}{b}$

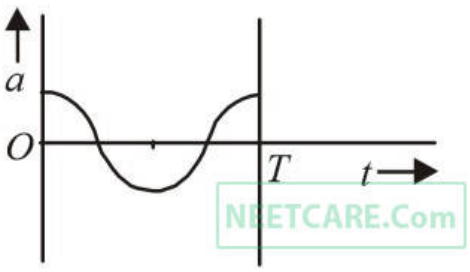
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Q30. AIPMT 2014

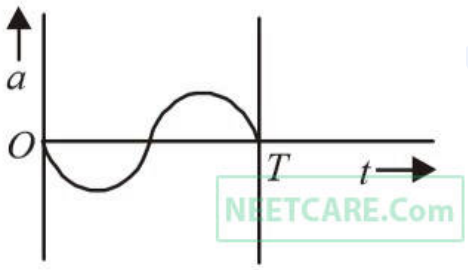
Open solution

The oscillation of a body on a smooth horizontal surface is represented by the equation,
X = A cos(ωt)
where X = displacement at time t
ω = frequency of oscillation
Which one of the following graphs shows correctly the variation of a with t?
Here a = acceleration at time t
T = time period

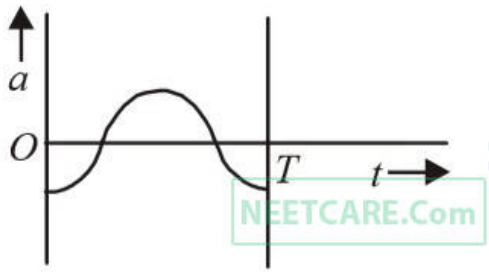
(A)



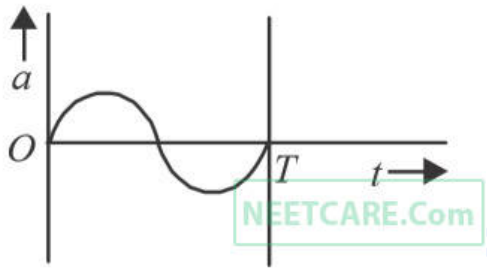
(B)



(C)



(D)



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Q31. NEET 2013 (Karnataka)

A particle of mass m oscillates along x-axis according to equation x = asinωt. The nature of the graph between momentum and displacement of the particle is

(A) Circle
(B) Hyperbola
(C) Ellipse
(D) Straight line passing through origin

Open solution

Q32. AIPMT 2011 Mains

Two particles are oscillating along two close parallel straight lines side by side, with the same frequency and amplitudes. They pass each other, moving in opposite directions when their displacement is half of the amplitude. The mean positions of the two particles lie on a straight line perpendicular to the paths of the two particles. The phase difference is

(A) $\frac{\pi}{6}$ (B) 0 (C) $\frac{2\pi}{3}$ (D) π

Open solution

Q33. AIPMT 2011 Prelims

Out of the following functions representing motion of a particle which represents SHM
(1) y = sinωt $-$ cosωt
(2) y = $\sin^3\omega$t
(3) y = 5cos$\left(\frac{3\pi}{4} - 3\omega t\right)$
(4) y = 1 + ωt + ω^2t^2

(A) Only (1)
(B) Only (4) does not represent SHM
(C) Only (1) and (3)
(D) Only (1) and (2)

Open solution

Q34. AIPMT 2010 Prelims

The displacement of a particle along the x-axis is given by x = a$\sin^2\omega$t. The motion of the particle corresponds to

(A) simple harmonic motion of frequency ω/π
(B) simple harmonic motion of frequency $3\omega/2\pi$
(C) non simple harmonic motion
(D) simple harmonic motion of frequency $\omega/2\pi$

Open solution

Q35. AIPMT 2010 Prelims

The period of oscillation of a mass M suspended from a spring of negligible mass is T. If along with it another mass M is also suspended, the period of oscillation will now be

(A) T (B) $\frac{T}{\sqrt{2}}$ (C) 2T (D) $\sqrt{2}T$

Open solution

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Q36. AIPMT 2009

A simple pendulum performs simple harmonic motion about x = 0 with an amplitude a and time period T. The speed of the pendulum at x = a/2 will be

(A) $\frac{\pi a}{T}$ (B) $\frac{3\pi^2 a}{T}$ (C) $\frac{\pi a\sqrt{3}}{T}$ (D) $\frac{\pi a\sqrt{3}}{2T}$

Open solution

Q37. AIPMT 2009

Which one of the following equations of motion represents simple harmonic motion ?

where k, k_0, k_1 and a are all positive.

(A) Acceleration = $-$ k (x + a)

(B) Acceleration = k(x + a)

(C) Acceleration = $-\omega^2 x$

(D) Acceleration = $-k_0 x + k_1 x^2$

Open solution

Q38. AIPMT 2008

Two simple harmonic motions of angular frequency 100 and 1000 rad s^{-1} have the same displacement amplitude. The ratio of their maximum acceleration is

(A) $1 : 10^3$ (B) $1 : 10^4$ (C) $1 : 10$ (D) $1 : 10^2$

Open solution

Q39. AIPMT 2007

A particle executes simple harmonic oscillation with an amplitude a. The period of oscillation is T. The minimum time taken by the particle to travel half of the amplitude from the equilibrium position is

(A) T/8 (B) T/12 (C) T/2 (D) T/4

Open solution

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Q40. AIPMT 2007

A mass of 2.0 kg is put on a flat pan attached to a vertical spring fixed on the ground as shown in the figure. The mass of the spring and the pan is negligible.

When pressed slightly and released the mass executes a simple harmonic motion. The spring constant is 200 N/m. What should be the minimum amplitude of the motion so that the mass gets detached from the pan (take g = 10 m/s^2).



Open solution

(A) 10.0 cm
(B) any value less than 12.0 cm
(C) 4.0 cm
(D) 8.0 cm

Q41. AIPMT 2007

The particle executing simple harmonic motion has a kinetic energy $K_0\cos^2\omega t$. The maximum values of the potential energy and the total energy are respectively

(A) $K_0/2$ and K_0
(B) K_0 and $2K_0$
(C) K_0 and K_0
(D) 0 and $2K_0$

Open solution

Q42. AIPMT 2007

The phase difference between the instantaneous velocity and acceleration of a particle executing simple harmonic motion is

(A) π
(B) 0.707 π
(C) zero
(D) 0.5 π

Open solution

Q43. AIPMT 2006

A rectangular block of mass m and area of cross-section A floats in a liquid of density ρ. If it is given a small vertical displacement from equilibrium it undergoes with a time period T, then

(A) $T \propto \frac{1}{\sqrt{m}}$
(B) $T \propto \sqrt{\rho}$
(C) $T \propto \frac{1}{\sqrt{A}}$
(D) $T \propto \frac{1}{\rho}$

Open solution

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Q44. AIPMT 2005

The circular motion of a particle with constant speed is

(A) periodic but not simple harmonic
(B) simple harmonic but not periodic
(C) period and simple harmonic
(D) neither periodic not simple harmonic.

Open solution

Q45. AIPMT 2004

A particle executing simple harmonic motion of amplitude 5 cm has maximum speed of 31.4 cm/s. The frequency of its oscillation is

(A) 4 Hz (B) 3 Hz (C) 2 Hz (D) 1 Hz

Open solution

Q46. AIPMT 2004

Two springs of spring constants k_1 and k_2 are joined in series. The effective spring constant of the combination is given by

(A) $\sqrt{k_1 k_2}$ (B) $(k_1 + k_2)/2$ (C) $k_1 + k_2$ (D) $k_1k_2/(k_1 + k_2)$

Open solution

Q47. AIPMT 2003

Which one of the following statements is true for the speed v and the acceleration a of a particle executing simple harmonic motion?

(A) When v is maximum, a is maximum
(B) Value of a is zero, whatever may be the value of v.
(C) When v is zero, a is zero
(D) When v is maximum, a is zero.

Open solution

Q48. AIPMT 2003

The potential energy of a simple harmonic oscillator when the particle is half way to its end point is

(A) $\frac{2}{3}$E (B) $\frac{1}{8}$E (C) $\frac{1}{4}$E (D) $\frac{1}{2}$E

Open solution

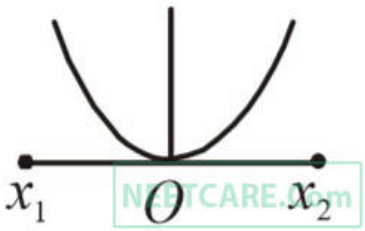
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Q49. AIPMT 2003

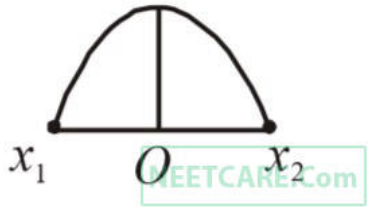
A particle of mass m oscillates with simple harmonic motion between points x_1 and x_2, the equilibrium position being O. Its potential energy is plotted. It will be as given below in the graph

Open solution

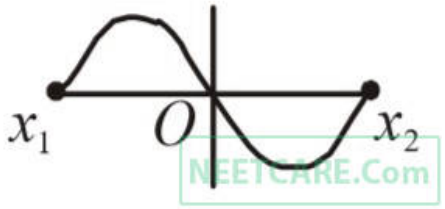
(A)



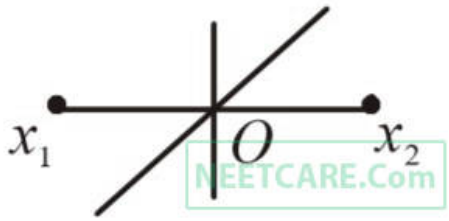
(B)



(C)



(D)



Q50. AIPMT 2003

The time period of mass suspended from a spring is T. If the spring is cut into four equal parts and the same mass is suspended from one of the parts, then the new time period will be

Open solution

(A) T/4 (B) T (C) T/2 (D) 2T

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Q51. AIPMT 2003

In case of a forced vibration, the resonance peak becomes very sharp when the

(A) damping force is small
(B) restoring force is small
(C) applied periodic force is small
(D) quality factor is small

Open solution

Q52. AIPMT 2002

Displacement between maximum potential energy position and maximum kinetic energy position for a particle executing simple harmonic motion is

(A) $\pm a/2$
(B) $+a$
(C) $\pm a$
(D) -1

Open solution

Q53. AIPMT 2002

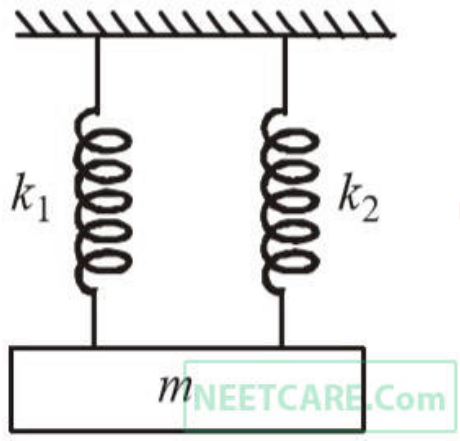
When an oscillator completes 100 oscillations its amplitude is reduced to $\frac{1}{3}$ of initial value. What will be its amplitude, when it completes 200 oscillations ?

(A) $\frac{1}{8}$
(B) $\frac{2}{3}$
(C) $\frac{1}{6}$
(D) $\frac{1}{9}$

Open solution

Q54. AIPMT 2002

A mass is suspended separately by two different springs in (successive order then time periods is t_1 and t_2 respectively, If it is connected by both spring as shown in figure then time period is t_0 , the correct relation is



(A) $t_0^2 = t_1^2 + t_2^2$
(B) $t_0^{-2} = t_1^{-2} + t_2^{-2}$
(C) $t_0^{-1} = t_1^{-1} + t_2^{-1}$
(D) $t_0 = t_1 + t_2$

Open solution

Q55. AIPMT 2001

The total energy of particle performing SHM depend on

(A) k, a, m
(B) k, a
(C) k, a, x
(D) k, x

Open solution

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Q56. AIPMT 2000

Two masses M_A and M_B are hung from two strings of length l_A and l_B respectively. They are executing SHM with frequency relation $f_A = 2f_B$, then relation

Open solution

(A) $l_A = \frac{l_B}{4}$ does not depend on mass

(B) $l_A = 4l_B$, does not depend on mass

(C) $l_A = 2l_B$ and $M_A = 2M_B$

(D) $l_A = \frac{l_B}{2}$ and $M_A = \frac{M_B}{2}$.

Q57. AIPMT 2000

The bob of simple pendulum having length l, is displaced from mean position to an angular position q with respect to vertical. If it is released, then velocity of bob at equilibrium position

Open solution

(A) $\sqrt{2gl\,(1-\cos\theta)}$

(B) $\sqrt{2gl\,(1+\cos\theta)}$

(C) $\sqrt{2gl\cos\theta}$

(D) $\sqrt{2gl}$

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Answer Key

1. **A**	2. **C**	3. **C**	4. **B**	5. **D**
6. **D**	7. **B**	8. **C**	9. **D**	10. **B**
11. **B**	12. **C**	13. **C**	14. **C**	15. **B**
16. **A**	17. **C**	18. **A**	19. **D**	20. **C**
21. **D**	22. **A**	23. **B**	24. **B**	25. **B**
26. **D**	27. **B**	28. **D**	29. **A**	30. **C**
31. **C**	32. **C**	33. **C**	34. **C**	35. **D**
36. **C**	37. **C**	38. **D**	39. **B**	40. **A**
41. **C**	42. **D**	43. **C**	44. **A**	45. **D**
46. **D**	47. **D**	48. **C**	49. **A**	50. **C**
51. **A**	52. **C**	53. **D**	54. **B**	55. **B**
56. **A**	57. **A**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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